

Sai Siddharth Cilamkoti

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SUMMARY

AI Software Engineer who builds production agentic LLM systems and backend systems end to end with problem framing, architecture, and delivery. Primary stack: Python, Java, FastAPI, Spring Boot, LangGraph, Postgres, Docker and Kubernetes on AWS, with multi-provider LLM integration. I work close to the people who use what I build, turning ambiguous user problems into shipped product. An agent system is only as trustworthy as what you can measure about it, so I build tracing, execution guardrails, and regression evals from day one. I can take full ownership of a system from first conversation to production and early-stage is where I want to do it next: a company being made from the ground up, where you own a piece of everything and wear whatever hat the problem demands.

EDUCATION

University at Buffalo

Master of Science in Computer Science

Buffalo, NY

Aug. 2023 -- Feb 2025

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science with AI

India

Aug. 2019 -- May 2023

EXPERIENCE

Associate AI Engineer

Nov 2025 -- Present

Midwest Partners LLC

Hibbing, MN (Remote)

- Owned the architecture and delivery of a production agentic NL-to-SQL platform over a hospitality business dataset. Chose a routed multi-path design orchestrated using LangGraph over a single prompt input and generation after evals showed 80% failure on schema heavy queries. Shipped to business managers and stakeholders to surface the insights behind their competitive positioning.
- Designed a Graph RAG schema layer linking tables and foreign-key relationships to keep generation grounded in the true schema, and built the reliability stack around it with sqlglot AST-level validation, read-only execution guardrails, self-correction retries, and narration constrained to retrieved rows, instrumented with per-stage Langfuse tracing, audit trails, and a regression eval harness; reached 90% path-selection accuracy and a 0.88 narration faithfulness score.
- Architected a competitor rate-intelligence pipeline (n8n, SerpAPI/Google Hotels, Supabase/Postgres) tracking competitors by running scheduled jobs, ingesting new reviews with sentiment scoring across 4 sources, alerting stakeholders via Slack and email on price and metasearch-ranking shifts that feed pricing decisions.

Research AI Engineer

Feb. 2024 -- Aug 2025

University at Buffalo | Co-authored a [paper](#) on this study, accepted at FLLM 2025

Buffalo, NY

- Designed and built a large-scale LLM evaluation pipeline benchmarking 20 frontier models (GPT-4o, Claude 3.5 Sonnet, DeepSeek-v3, Gemini, Llama, Grok) across 878 coding tasks with 100+ test cases each, using OpenAI, Anthropic, Google Vertex AI (GCP), and AWS Bedrock APIs with batch inference, and iterative prompt-engineering feedback loops that lifted Pass@25 from 51% to 92%.
- Built an automated, agentic LLM-as-judge review pipeline with multi-agent orchestration that scored generated solutions and ran 25-iteration regeneration on execution errors, producing evaluation quality signals that flagged DeepSeek-v3 and GPT-4o as most efficient while Gemini-1.5-Pro ran up to 450x less efficiently on Dynamic Programming and Backtracking tasks.
- Optimized a transformer-based model (GPT-2) for text summarization via knowledge distillation, 8-bit quantization, and structured pruning, cutting inference latency and energy consumption by 30% while retaining 95%+ of baseline ROUGE quality on the evaluation benchmark.

Software Engineer

Aug. 2024 -- Dec. 2024

Kaleida Health

Buffalo, NY

- Owned end-to-end feature delivery across a React frontend and Java/Spring Boot distributed microservices, containerizing services with Docker and deploying to AWS via Kubernetes (EKS) with CI/CD pipelines, serving users across 7 departments.
- Ran weekly design and requirements sessions with hospital doctors and admin staff, translating their workflow constraints into the access model: patient-delegated, role-scoped visibility across patient, family, doctor, and admin roles, backed by audit trails and secured with JWT, HTTPS, and RBAC. Iterating with them directly surfaced the delegation requirement the original spec had missed.
- Built real-time WebSocket messaging (1:1 and group) integrated in to the platform with auto-translation via the Google Translate API sustaining 100s of concurrent connections, and drove backend API design, SDLC, and documentation across cross-functional teams to ship the stakeholder-demoed platform in 16 weeks.

PROJECTS

Exergy - Founding Engineer | Python, FastAPI, NextJS, Supabase

[Link](#) | 2024

- Built a two-sided solar marketplace from zero to first iteration, owning the backend architecture and data model (FastAPI, Supabase/Postgres) alongside assisting with the Next.js/React frontend development.
- Designed the sizing and economics engine that turned a homeowner's address, roof area, and uploaded utility bill into a recommended system size, projected generation, investment, savings, and payback period.
- Built a privacy-controlled quote flow letting users solicit competing bids from up to 5 vendors while disclosing only chosen fields, with responses compared side by side in-platform. Partnered with the sales team on live homeowner and retailer feedback, turning it into shipped product changes.

InlineAsk - Browser Extension | TypeScript, Chrome MV3, Anthropic & OpenAI APIs

[GitHub](#) | 2026

- Built and shipped an end-to-end Chrome extension in TypeScript enabling inline LLM follow-up questions on highlighted text across

Claude, ChatGPT, Gemini, and Copilot, with a pluggable multi-provider layer integrating the Anthropic and OpenAI APIs.

- Engineered frontend DOM adapters applying context-engineering discipline — retrieval-style context grounding under a strict 4000-token budget — with AES-GCM credential encryption and retry/timeout handling to harden edge cases and ensure reliability.

TECHNICAL SKILLS

Programming: Python, Java, SQL, Javascript, C++, HTML/CSS

AI & ML: PyTorch, LangChain, LangGraph, VectorDBs, Huggingface, Cohere Rerank, Fine-Tuning, Prompt Engineering, RAG, LangFuse, LangSmith, Google ADK, Crew AI, AWS Bedrock, Gemini, Anthropic, OpenAI API, Numpy

Backend & API: Spring Boot, FastAPI, Flask, WebSockets, Microservices, Postgres, Supabase

DevOps and Tools: Docker, Kubernetes, GitHub Actions, AWS, Azure Dev Tools, Git, Linux, Cursor, Claude Code